

COURSE SYLLABUS: CS 480 - ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Semester: Fall 2026

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Course Overview:

This course provides a comprehensive mathematical and practical foundation in modern artificial intelligence, machine learning algorithms, deep learning representations, and autonomous decision systems.

Unit 1: Foundations of Supervised Learning

1.1 Linear Regression and Gradient Descent

Formulation of empirical risk minimization, mean squared error loss, analytical normal equations versus iterative stochastic gradient descent optimization.

1.2 Logistic Regression and Classification Metrics

Binary cross-entropy loss, sigmoid activation, decision boundaries, ROC curves, AUC, precision, recall, and F1-score evaluation.

1.3 Regularization and the Bias-Variance Tradeoff

L1 Lasso and L2 Ridge regularization penalties, model capacity, overfitting, underfitting, and cross-validation methodologies.

Unit 2: Deep Learning Architectures

2.1 Deep Neural Networks and Backpropagation

Multi-layer perceptrons, forward propagation, computational graphs, reverse-mode automatic differentiation, gradient vanish and explosion challenges.

2.2 Convolutional Neural Networks for Vision

Spatial invariance, convolution kernels, stride, padding, pooling layers, ResNet skip connections, feature map hierarchies.

2.3 Sequence Models and Transformer Attention

Recurrent neural networks, LSTM gating mechanisms, self-attention equations, multi-head scaled dot-product attention, positional encodings.

Unit 3: Unsupervised Learning & Dimensionality Reduction

3.1 Clustering Algorithms

K-means clustering algorithm, centroid initialization heuristics, inertia, and hierarchical agglomerative clustering.

3.2 Dimensionality Reduction

Principal Component Analysis (PCA), singular value decomposition (SVD), variance explained ratio, and t-SNE embeddings.

Unit 4: Reinforcement Learning & Autonomous Agents

4.1 Markov Decision Processes (MDPs)

States, actions, transition probabilities, discount factors, reward functions, and Bellman optimality equations.

4.2 Value-Based and Policy-Based Algorithms

Q-learning, deep Q-networks (DQN), exploration versus exploitation strategies (epsilon-greedy), and policy gradient theorem.